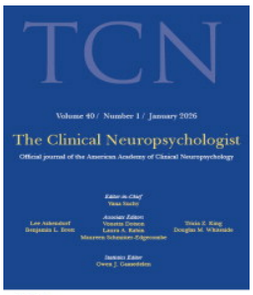

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Artificial intelligence and natural language processing in modern clinical neuropsychology: A narrative review

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ABSTRACT

Objectives: Advances in natural language processing (NLP) promise to augment traditional neuropsychological assessment by transforming speech and text into objective digital biomarkers. This narrative review synthesizes NLP research, evaluates its incremental diagnostic value across neurodegenerative, neurological, neurodevelopmental and psychiatric disorders, and posits recommendations for adoption in clinical neuropsychology. **Methods:** A scoping search of PubMed, Embase, PsycINFO, Scopus and Web of Science retrieved 56 empirical studies applying NLP within neuropsychological contexts. Manuscripts were critically appraised with attention to data source, linguistic features, modelling approach, validation strategy and clinical utility. **Results:** Across neuropsychological syndromes, NLP reliably extracts lexical, syntactic and acoustic markers with pooled area-under-the-curve estimates exceeding 0.85, often outperforming legacy tests while requiring only brief speech samples or existing electronic health-record text. Transformer-based language models further enable real-time documentation support, longitudinal surveillance and personalized feedback. Nonetheless, small homogeneous training sets, limited external calibration and opaque decision pathways threaten generalizability and clinician trust, and implementation of NLP must address algorithmic bias, cultural-linguistic representativeness, ethical privacy standards, and explainability. **Conclusions:** To realize NLP's potential, neuropsychologists must cultivate foundational literacy in computational linguistics, follow transparent reporting, embed privacy-preserving pipelines, and co-design explainable dashboards that contextualize machine inferences within holistic case formulations. Scaled, demographically balanced consortia and multimodal fusion with neuroimaging and wearables are priority directions. Properly implemented, NLP can render assessment more objective, efficient and equitable, positioning language as a central biomarker and integrating linguistically informed artificial intelligence to extend the reach of neuropsychological services.

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Introduction

While neuropsychological services are integral to the diagnosis and management of neurological conditions, the field has experienced limited technological innovation over the past century. The continued reliance on traditional, non-digitized assessment methods stands in contrast to the technological advancements seen in other medical disciplines. The emergence of novel digital health technologies, including computerized cognitive testing, ecological momentary assessment, and various sensing modalities, presents an opportunity to enhance neuropsychological practice and patient outcomes (Harris et al., 2024). While traditional assessment methods possess robust psychometric validity, they are often constrained by subjective interpretation, clinician variability, resource and time demands, limited scalability, and reduced criterion validity (Bilder & Reise, 2019). Advancements in artificial intelligence (AI), and specifically natural language processing (NLP), offer a means of forward progress, by automating the interpretation and analysis of spoken or written data generated during clinical interactions. Artificial intelligence more broadly, although not NLP specifically, has previously been reviewed (Graham et al., 2020). NLP applies computational linguistic techniques to automate the extraction and analysis of speech and textual data, aiming to enhance objectivity, diagnostic precision, and efficiency in clinical practice (Clay et al., 2025).

NLP is defined as a suite of computational techniques facilitating the analysis and representation of naturally occurring text at various linguistic levels (Crema et al., 2022). An NLP marker is conceptualized as a "digitally acquired, computationally derived, quantifiable measure or set of measures of human language production reflecting the state of biological, neurocognitive, and social processes that contribute to it" (Corona Hernández et al., 2023). Such markers can serve descriptive roles (e.g. for screening, trial stratification, outcome prediction) or mechanistic roles (e.g. indicative of pathologies at cellular, physiological, or circuit levels, potentially improving prediction accuracy and monitoring treatment response). NLP tasks can be extractive (e.g. named entity recognition, relation extraction, question answering, topic modeling, text similarity, sentiment analysis) or generative (e.g. translation, summarization, coherent text generation as in large language models).

Language is a critical neurocognitive system, wherein alterations often reflect underlying pathology when cortical integration is disrupted, ranging from psychopathologies such as schizophrenia to neurodegenerative syndromes such as primary progressive aphasias (Hinzen & Palaniyappan, 2024). With NLP, verbal productions can be converted into digital biomarkers, being objective, quantifiable behavioral data collected *via* digital devices, facilitating low-cost detection, classification, and monitoring of pathology which enables cognitive assessment across these syndromes and disease transitions (Gagliardi, 2024).

By providing objective, granular data on linguistic features, NLP can supplement subjective clinical observations, offering a more nuanced understanding of cognitive function and pathology. Given the increasing interest in NLP within neuropsychology, this narrative review summarizes the literature, evaluates the utility of NLP across various clinical conditions, and provides considerations for practical implementation

within neuropsychological contexts, highlighting how these tools can empower neuropsychologists and shape the future of the profession.

Objectives

This review had three objectives:

1. To provide a technical overview of natural language processing approaches to data, including collection methods, data elements utilized, purpose of analysis, and analytical techniques.
2. To synthesize current applications of NLP within specific populations relevant to clinical neuropsychology, illustrated through use cases in neurodegenerative, neurological, neurodevelopmental, and psychiatric syndromes.
3. To propose recommendations addressing challenges and limitations, and future directions for the adoption of NLP in clinical neuropsychology.

Methods

This review is intended as a conceptual and narrative overview rather than a quantitative systematic review of empirical studies. Searches were conducted across PubMed, Embase, PsycINFO, Scopus, and Web of Science databases, without restrictions on publication date or study design (see Appendix A). English-language articles applying NLP within clinical neuropsychological contexts were considered. Searching for the specific terms with inclusion of "neuropsychology" returned only 56 publications, while an expanded search produced 122 original manuscripts relevant to other medical fields (e.g. neurology, psychiatry). Representative and exemplar studies were then selected to illustrate the application of specific techniques, summarize the current state of the field, and identify relevant directions for clinical neuropsychology.

Technical overview of NLP approaches to data analysis and interpretation

Across medicine, NLP applications typically are employed across three key areas: information extraction (applying systems to extract information from patient records for insights); classification (using NLP features to train models predicting patient health status); and outcome prediction (using NLP tools to form hypotheses, such as stratifying candidates for medical procedures; Crema et al., 2022). NLP studies use diverse data sources and a range of extraction methods, from clinical notes, structured cognitive assessments, electronic health records (EHRs), audio recordings of patient interviews, and genomic data. Steps in analysis, including current limitations, are outlined below. Table 1 includes a glossary of terms and definitions.

Data collection and preprocessing

Data collection frequently involves recording and transcribing patient interviews and clinical interactions into textual data suitable for analysis. This data undergoes

Table 1. Glossary of technical terms.

Term	Definition
Artificial Intelligence	A field of computer science dedicated to the creation of systems, known as intelligent agents, that can perceive their environment and take actions to achieve specific goals. It encompasses a spectrum of theories and technologies aimed at simulating cognitive functions typically associated with the human mind, including reasoning, knowledge representation, planning, learning, natural language processing, perception, and the ability to manipulate objects.
Attention	A neural mechanism that dynamically places greater weight on the most relevant portions of the surrounding text, regardless of how far those portions are from the word being processed. It enables the model to give extra emphasis to parts of the input it deems most relevant for producing the output.
Augmentation Strategy	Any method that fabricates extra, artificial training examples to enlarge and diversify the original data set.
Autoregression	A statistical approach in which future values of a sequence are predicted solely from preceding values, fundamental in time-series modelling.
Bidirectional Encoder Representations from Transformers	BERT is a Transformer-based model trained with masked-language-modeling and next-sentence objectives so it can read input text in both directions, leading to state-of-the-art results on tasks ranging from question answering to sentiment analysis.
Corpus/Corpora	A body of written material (structured or unstructured, annotated or raw) assembled for linguistic or computational study.
Deep Learning	A subset of machine learning that trains multi-layer neural networks whose stacked layers progressively learn more abstract representations of data.
ELMo (Embeddings from Language Models)	A deep, bi-directional LSTM system from the Allen Institute that produces context-dependent word vectors, capturing both syntax and semantics; the original version contains roughly 94 million parameters and boosts tasks such as sentiment analysis, NER, and text classification.
Entity Linking	The task of connecting recognized entities in text with their corresponding entries, concepts, or identifiers in an external knowledge base.
Feed-forward Network	A neural network where information flows from input to output through successive layers without cycles or feedback connections.
Fine-tuning	Continued training of an already pre-trained model on labelled, task-specific data so the model specializes in a narrower domain.
GPT (Generative Pre-trained Transformer)	OpenAI's family of Transformer models trained on vast text corpora to predict the next token; parameter counts rose from 117 million in GPT-1 to 1.5 billion in GPT-2 and 175 billion in GPT-3, enabling strong performance in generation, dialogue, and many other tasks after fine-tuning.
Labeling	The act of attaching descriptive tags or values to raw data instances so they can serve as targets in supervised learning.
Lemmatization and Stemming	Pre-processing steps that reduce words to a base form: stemming heuristically chops affixes, which can produce non-words, while lemmatization consults vocabulary rules to return the correct dictionary form.
Long Short-Term Memory System (LSTM)	An advanced recurrent neural network (RNN) architecture designed to overcome the vanishing gradient problem that limits the ability of traditional RNNs to learn long-range dependencies. This is achieved through a more complex repeating module that incorporates a dedicated memory cell and three primary gating mechanisms.
Machine Learning (ML)	A subfield of AI that employs statistical algorithms to enable computer systems to iteratively learn from data, identify complex non-linear patterns, and make data-driven predictions or decisions without explicit programming. Models can automatically improve their performance on a given task as they are exposed to more data. These can be supervised (an algorithm trained on a labeled dataset where input variables are paired with correct outputs), or unsupervised (an algorithm explores an unlabeled dataset to identify inherent structures, clusters or anomalies within the data).
Named Entity Recognition (NER)	A system that spots and labels spans of text that belong to predefined semantic classes, such as people, organizations, or locations, providing a basis for many downstream tasks.
Natural Language Processing (NLP)	A subfield of artificial intelligence and computational linguistics dedicated to equipping computers with the ability to process, analyze, and interpret written or spoken human language data.
Part-of-speech tagging	The procedure that assigns a grammatical category (e.g. noun, verb, adjective) to every token in a sentence.

(*Continued*)

Table 1. Continued.

Term	Definition
Pre-training	An initial learning phase where a model derives broad linguistic patterns from large amounts of unlabeled text before any task-specific adjustment.
Question Answering (QA)	Given a passage and a query, the model pinpoints the portion of text that answers the question and can also recognize when no answer is present.
Reinforcement Learning	A learning framework in which an agent improves its behavior through trial-and-error interactions with an environment, guided by reward signals.
Relation Extraction	The process of identifying and categorizing the semantic link that joins two referenced entities within the same sentence.
Sentiment Analysis	A method that determines whether a passage conveys positive, neutral, or negative emotional tone.
Summarization	Condensing a document while retaining its key information.
Text Generation	Producing coherent, human-readable sentences or documents from input data, for example powering chatbots or creating extra training examples.
Text Similarity	A procedure that yields a numerical score indicating how closely the meanings of two pieces of text align.
Tokenizer	The component in an NLP pipeline that breaks text into smaller units (words, sub-words, or characters) known as tokens.
Topic Modeling	A technique that uncovers the principal themes recurring throughout a collection of documents.
Transfer Learning	A paradigm in which knowledge gained while solving one problem is carried over to speed up or improve learning on a different, related task.
Transformer	A neural-network architecture built wholly around self-attention, enabling the model to capture long-range relationships across an input sequence and its output. It is first devised for text, that simultaneously attends to all tokens in a sequence *via* self-attention, greatly improving language-task performance.
Translation	Rendering text from one language into another, the earliest classical application of NLP.

preprocessing, a step encompassing techniques such as tokenization (segmenting text into tokens), case normalization (converting text to lowercase), lemmatization and stemming (reducing words to root forms, e.g. "machinery," "machinations" to "machin"), part-of-speech tagging (categorizing words grammatically), and removal of stop words (irrelevant or redundant words; Clay et al., 2025). Anonymization is also necessary for unstructured text commonly found in EHRs. This step aims to balance recognition accuracy and speed, with some models attaining excellent speech to text conversion accuracy (e.g. precision 0.73, recall 0.90). However, pre-processing method choice can impact accuracy by up to 25%, and performance is often improved by human editing, highlighting limitations with the current technology (Ng et al., 2025).

Analytical methodologies

NLP analysis primarily employs two types of methods: statistical and artificial neural network (ANN)-based approaches. Statistical methods derive insights based on quantitative measures of text content, with techniques such as Bag of Words (BoW), an encoding strategy analyzing word frequencies, and Term Frequency-Inverse Document Frequency (TF-IDF), which weights terms based on their predicted impact. ANNs are interconnected systems of "nodes" that process information similarly to neural networks in the brain. The strength of connections and output between layers are dynamically modified through training data, allowing the network to "learn" patterns *via* weighting (Clay et al., 2025). In ANNs, "embedded" refers to assessing relationships between words or phrases and adjacent terms. There are also convolutional neural networks (CNNs) and recurrent neural networks (RNNs), often applied to NLP tasks, and Long

Short-Term Memory networks (LSTM), a type of RNN capable of retaining information over longer periods to address the challenge of learning long-term temporal dependencies (Crema et al., 2022).

Large language models

Large Language Models (LLMs) are AI systems using generative NLP, trained on large text datasets capable of generating human-like language from contextual prompts. They use ANN architectures, often Transformers, pre-trained on broad corpora (often on the internet) to learn document features and develop confidence in generating plausible outputs for unseen data. Recent advancements favor advanced language models employing transfer learning, such as Bidirectional Encoder Representations from Transformers (BERT) and Generative Pre-trained Transformer (GPT) models (Crema et al., 2022). BERT excels in contextual understanding and has been applied to clinical text for tasks such as named entity recognition (NER), sentiment analysis, and clinical note classification. Clinical BERT and Clinical BioBERT are models fine-tuned on medical corpora, demonstrating the utility of domain-specific contextual embeddings for biomedical data, which could be applied in neuropsychological research and assessment (Crema et al., 2022). These neural networks are organized into layers, each with multiple attention heads and feedforward mechanisms. Attention heads determine which parts of the input text the model focuses on, aiding in understanding context and word relationships (Romano et al., 2023). LLM training involves predicting the next word, filling masked words, or generating new sentences; the model learns patterns by minimizing discrepancies between predicted and actual outputs. LLMs can function bidirectionally, predicting words from both left and right contexts, which is valuable for analyzing spoken language, such as filling gaps in a patient's narrative due to memory loss or aphasia. LLMs can detect subtle linguistic changes in word production, vocabulary, sentence structure, or syntax simplification, potentially earlier than clinicians or patients. They can also be applied in rehabilitation, learning from data on a patient's linguistic abilities to generate personalized word games or story activities, track performance, adjust difficulty, and provide dynamic cognitive stimulation (e.g. semantic and phonemic fluency tasks with metrics like word count, correctness, and temporal variation).

NLP tasks and pipelines

NLP analysis pipelines typically follow structured steps: data transcription, preprocessing, feature extraction, modeling, and interpretation and visualization of outcomes. The most accessible tools currently available are standardized software libraries, such as Python's Natural Language Toolkit (NLTK), spaCy, Hugging Face, and Scikit-learn, support these analyses and promote reproducibility (Clay et al., 2025). NLP tasks executed within clinical neuropsychology include NER, relation extraction, sentiment analysis, question-answering systems, topic modeling, text summarization, and translation. NER automates the extraction of structured clinical information from unstructured text, identifying entities including diagnoses, symptoms, medications, and treatment plans, thereby streamlining documentation. Sentiment analysis and topic modeling extract emotional and thematic content from patient interactions, providing

insights into patient experiences and facilitating personalized interventions (Crema et al., 2022).

For NLP tasks in EHRs, applications can be divided into information extraction and classification or prediction. Information extraction focuses on objective parameters (diagnoses, medications, neuroimaging procedures) using NER, or subjective data (emotional content) using sentiment analysis. Classification and prediction models frequently employ machine learning models such as Random Forests, Support Vector Machines (SVMs), and various neural network architectures to analyze linguistic features. Classification tasks depend on training data: supervised approaches use labeled data, while unsupervised approaches classify data based on features identified without human input (Clay et al., 2025).

Performance metrics

Internal validation commonly employs k-fold cross-validation or nested cross-validation to mitigate overfitting. External validation against independent clinical cohorts or across languages remains less common and is a current limitation on generalizability. Performance is typically reported using discrimination indices such as sensitivity, specificity, and area-under-the-receiver-operating-characteristic curve (AUC). Calibration measures (e.g. Brier score, calibration-in-the-large) are often omitted, limiting direct comparison with classical psychometric indices. Nevertheless, the limited reviews available indicate adequate performance metrics, with models achieving good sensitivity (median 0.88, interquartile range [IQR] 0.74–0.91) and specificity (median 0.96, IQR 0.81–0.99) in detecting cognitive impairment against reference standards such as cognitive testing, validated criteria, objective biomarkers, and diagnostic codes (Shankar et al., 2025). These metrics suggest that NLP methodologies could enhance the diagnostic accuracy and clinical utility of neuropsychological assessments, using linguistic features from speech transcripts and clinical documentation to differentiate conditions relevant to clinical neuropsychology with good sensitivity and specificity.

Synthesis of current NLP applications in clinical neuropsychology

Neurodegenerative disorders

Research on dementia-related language biomarkers typically follows a workflow involving the collection and annotation of language samples, automated extraction of quantitative linguistic indices, and training and validation of machine-learning classifiers. This approach aims to detect communicative impairment for screening or diagnosis and track linguistic changes over time, broadly categorized into three classes: speech-based features (e.g. acoustical measures, rhythmic indices), text-based features (e.g. lexical richness, syntactic complexity, semantic structure, pragmatic attributes), and extra-linguistic or multimodal features (e.g. gaze, facial expression; Gagliardi, 2024).

As exemplar studies in detection, the Framingham Heart Study has demonstrated that automated transcription of voice recordings from neuropsychological tests (Logical Memory, Paired Associated Learning, Digit Span, Similarities, Verbal and Category Fluency) can differentiate normal cognition from dementia and mild

cognitive impairment (MCI), with AUCs ranging from 0.88 to 0.93 and can differentiate features for MCI to AD conversion over six years (voice to text NLP analysis AUC 77.8% compared to neuropsychological tests AUC 71.3%; Amini et al., 2023, 2024; Thomas et al., 2020). NLP has also shown utility in identifying neuropsychiatric symptoms in dementia from EHRs, achieving good internal validation (AUC 0.81–0.91), although external validation across cohorts was more variable (AUC 0.51–0.93). Apathy, anxiety, aberrant motor behavior, irritability, and depression were consistently identified in clinician-entered clinical EHR notes as key symptoms, although often poor agreement with reference standards (e.g. caregiver Neuropsychiatric Inventory; kappa range 0.49–0.74; independent clinician raters kappa range 0.92–0.35–0.99) suggesting rater differences may be important given reliance on proxy-based instruments in outpatient clinics and in clinical trials (Eikelboom et al., 2023). Data from Speech-in-Alzheimer's programs (AMYPRED, Alzheimer's Disease Neuroimaging Initiative-4 [ADNI4]-remote, ADNI4 in-clinic) using the automated Storyteller story-recall task demonstrated robust transferability of speech models for detecting prodromal AD. Text-based metrics yielded AUCs between 0.71 and 0.84 across ADNI4 samples (Skirrow et al., 2024). Subtle linguistic changes (lexical variety, repetition, indefinite terms, noun/pronoun ratios, syntax, pauses, speech rate) emerge early and can mirror amyloid/tau biomarker status. This early detection capability is promising for timely intervention and disease management.

Beyond classification, NLP reveals direct links between language and underlying neuropathology. Analysis of one-minute speech samples has differentiated cohorts and patient populations (AUC 0.88 for AD versus frontotemporal dementia, 0.93 for patients versus healthy controls) and identified anatomical correlates from structural magnetic resonance imaging (MRI), for instance, noun frequency and pause rate correlate with grey matter volume loss in the limbic and salience networks (Cho et al., 2024). Lexical–semantic models (AUC 0.80) and acoustic models (AUC 0.77) analyzing a picture description task using the standardized Geneva Minimalistic Acoustic Parameter Set (GeMAPS) approach outperform the Boston Naming Test (AUC 0.66) in separating MCI from healthy aging individuals, and can identify amyloid-β positivity. Specifically, acoustic metrics correlate with hippocampal volume (R^2 = 0.03, beta = −60.7, p = 0.017), while lexical–semantic metrics tracked CSF Aβ42 levels, and both predicted clinical decline over two years (beta = 0.26, p = 0.009; Hajjar et al., 2023). Hour-long interview transcripts achieve sensitivities and specificities around 82% when contrasting amyloid-negative cognitively unimpaired (CU) participants with AD patients, and autobiographical speech was the most diagnostically informative segment, suggesting task choice for NLP is important (Balabin et al., 2025).

In Parkinson's disease, NLP reviews indicate similar utility, although with greater heterogeneity (overall accuracy 43–94%, sensitivity 8–95%, specificity 3–100%, AUC 0.32–0.97). Lexico-semantic measures (e.g. verb-to-token proportions, noun rates, conjunctions, pronouns, negatives, word-frequency, prefix statistics) are commonly the most effective discriminators (40%), followed by NLP-derived composites (26%) and morphological consistency indices (20%; Palmirotta et al., 2024). Comparator studies have used AI-driven approaches to identify linguistic features including retracing, action verb, utterance errors, and verbless-utterance ratios that can differentiate

PD from mild cognitive impairment and healthy controls, implicating frontostriatal brain mechanisms such as reduced verbal self-monitoring (Aresta et al., 2025).

Synthesis of neurodegeneration research. For neuropsychologists, these findings demonstrate that NLP can provide efficient, sensitive, and biologically grounded markers for neurodegenerative diseases. The ability of a brief speech sample to outperform a standard naming test or to reflect a patient's amyloid status represents a significant potential shift in clinical assessment. The key value lies in augmenting traditional testing with objective data to improve differential diagnosis and track disease progression. Further, the efficiency of collecting a six to eight minute speech sample compared to multi-hour traditional batteries highlights NLP's practical advantages for busy clinical settings. However, the heterogeneity in performance metrics highlights a need for standardized methodologies and reporting, and the importance of choice of reference standards. Further, most studies are cross-sectional with a focus on detection, and longitudinal research is needed to validate these markers for monitoring change over time and predicting functional outcomes.

Neurological syndromes and injuries

NLP applications have emerging applications across other neurological syndromes. Brain tumor (glioblastoma, pituitary adenoma, meningioma) patients have been effectively identified from free-text EHR entries using the Systematized Nomenclature of Medicine—Clinical Terms (SNOMED-CT) ontology (Booker et al., 2025). In epilepsy services, NLP is used to categorize clinical documentation, extract structured information for diagnosis (International League Against Epilepsy classifications), differentiate epileptic from non-epileptic seizures using audio features and home videos, and analyze patient-generated online posts (Yew et al., 2023). Text mining has assigned focal or generalized seizure types for up to 92% of patients with previously unspecified epilepsies, provided psychoeducation, and domain-specific LLMs (e.g. Foresight) have forecasted medical outcomes by extracting longitudinal EHR data (Van Diessen et al., 2024). NLP has also accurately identified potential candidates for resective epilepsy surgery, with the capacity to generate dynamic candidacy scores over time and summarize clinical course (Wissel et al., 2020).

In traumatic brain injury (TBI), NLP is increasingly used to combine head computed tomography (CT) scans with clinical reports. Neuroimaging narrative reports are an under-exploited data source. BrainNERD, a two-stage NER system, flags lesion type, location, size, and chronicity from neuroradiologist reports, achieving precision and recall scores of 98–99% in internal and external validation, comparable to expert review (Torres-Lopez et al., 2022). This freely available tool can link imaging signatures to cognitive outcomes at scale, providing neuropsychologists with valuable objective data for understanding the impact of TBI.

In delirium, NLP pipelines analyzing unstructured EHR notes from in-patients flag delirium episodes, creating a curated lexicon of delirium-indicative terms with an area-under-the-receiver-operating-characteristic curve (AUROC) of 0.92, and free-text models performed slightly less well (AUROCs 0.826–0.830). Tokens including "confused," "delirium," "hallucinations," and "fluctuation" consistently ranked highest in feature

importance (Amjad et al., 2024), highlighting the potential for NLP to assist in the timely identification of this condition.

Synthesis of neurological syndrome research. In neurological syndromes, NLP applications are primarily emerging as tools for streamlining clinical workflows by extracting structured information from unstructured data. For neuropsychologists, the applications in these syndromes highlight a path toward greater efficiency and data integration. Tools that can automatically extract and structure information from EHRs or radiology reports can reduce administrative burden and provide objective, aggregated data to inform case formulations. The primary gap is the transition from retrospective validation to prospective, real-world implementation. While these tools show high accuracy, they are not yet seamlessly integrated into most clinical workflows for real-time decision support for clinicians.

Neurodevelopmental syndromes

Neurodevelopmental syndromes can benefit from NLP applications analyzing linguistic and communicative behaviors. Speech and language patterns (acoustic aspects, and language usage such as repetitiveness, idiosyncrasy, coherence) are effective for early detection and intervention planning in autism, although prosodic findings require larger studies for generalizability (Trayvick et al., 2024). In attention-deficit/hyperactivity disorder (ADHD), NLP analysis used in tandem with machine learning methods complements traditional assessments with diagnostic accuracy 70–95%, across questionnaires, physiological signals, MRI, virtual reality, accelerometers, actigraphy, ocular measures, and genetics (Zaheer & Akhtar, 2025). NLP methods with stylometric analysis of self-defining memories (measuring word count, lexical diversity, lexical density and cohesion) can distinguish adolescents with ADHD from controls, particularly delayed use of first-person pronoun (Barrios et al., 2025). NLP has also shown incremental value in rare disorders; in adults with phenylketonuria (PKU), NLP analysis of the Cookie Theft Task revealed significant differences on 23 linguistic features (attention to detail, clarity, cohesion, context, emotion recognition) despite no group differences on standard Weschler Adult Intelligence Scale (WAIS) subtests (Waisbren et al., 2025). This demonstrates NLP's sensitivity to subtle linguistic markers that may be missed by traditional tests.

NLP applied to EHRs can identify neurodevelopmental phenotypes, such as an autism phenotype ontology derived from 3336 terms linked to 1943 medical concepts for automated diagnosis and subtyping (Zhao et al., 2022). In Dravet Syndrome, EHR analysis distinguished the syndrome from febrile seizures earlier than typical diagnosis age by identifying concepts related to the DS phenotype and non-seizure concepts (neurodevelopment, gait disorders; Lo Barco et al., 2021). Using a publicly available speech database from children with specific language impairment, one study used a CNN deep learning framework combined with LTSM model to attain 100% accuracy in discriminating individuals with and without language disorder using raw speech signals (e.g. phonations of vowels "a" and "o"; Sharma & Singh, 2022). Further, using home-based video analysis of behavioral features (i.e. expressive language, echolalia, eye contact, spontaneous gestures, emotion expression, communicative engagement, responsiveness, comforting others, aggression, sharing of excitement) another study achieved autism classification accuracy close to 92% (Nabil et al., 2021).

Synthesis of neurodevelopmental syndrome research. These findings are relevant for neuropsychologists working with complex pediatric and adult neurodevelopmental populations. NLP provides objective, granular data to enhance differential diagnosis where traditional tests may lack sensitivity. The potential for earlier and more accessible diagnosis, such as through home video analysis, could improve access to care. However, a major gap is the need for generalizability. Prosodic findings in autism, for example, require larger studies to be confirmed. Further, while studies report high accuracy, many are restricted to specific datasets. Validation in larger, more diverse clinical populations is a critical next step.

Psychiatric disorders

NLP research in psychiatry focuses on detection (categorical diagnosis, symptom quantification, early warnings), monitoring (treatment response), prediction (suicide risk, relapse), and treatment selection (matching patients to interventions). Emotion-laden acoustic profiles differentiate depression subtypes, pause length tracks negative symptoms in youth at clinical high risk (CHR) for psychosis, and shifts in lexical choices can precede psychosis (Stanislawski et al., 2021). Monitoring applications track treatment response; speech-pause behavior and fundamental frequency distinguish responders in major depression. Prediction is active, with lexical patterns signaling aggression risk and linguistic changes forecasting first-episode psychosis or later schizophrenia diagnosis. Treatment selection suggests language can inform prescribing, for instance, higher baseline use of affect-laden words predicts favorable psilocybin response in depression (Carrillo et al., 2018).

In schizophrenia and depression, NLP methods accurately identify characteristic linguistic patterns, including reduced lexical diversity, semantic incoherence, and negative sentiment. These linguistic features facilitate improved monitoring of symptom severity, treatment response, and overall disease trajectory, significantly augmenting traditional assessment methods (Crema et al., 2022). Sentiment analysis and linguistic complexity measures have proven useful in detecting depressive symptomatology, predicting suicide risk, and monitoring treatment response (Malgaroli et al., 2023). EHR analysis has shown that NLP-based predictors improve the prognostic accuracy of psychosis risk calculators, and specifically, adding NLP-derived words (tearfulness, poor appetite, weight loss, insomnia, substance use terms, guilt, irritability, delusions, hopelessness, disturbed sleep, poor insight, agitation, paranoia) enhances early detection (Irving et al., 2021). Combining automatic speech recognition with semantic NLP models yields good classification accuracy and concordance with manual transcription (Ciampelli et al., 2023). Studies further indicate that NLP measures provide complementary information, and that the choice of speech generation method can impact results (e.g. speech generated from story retelling and Thematic Apperception Test [TAT] outperforms free speech tasks in first episode psychosis) (Morgan et al., 2021).

Linguistic features extracted from dyadic task transcripts in people with psychotic disorders predict current suicidal ideation, past attempts, and future suicidal behavior (F1 0.86–0.93). Top features varied: cognition-related (differentiation, tentative language, analytic thinking, cognitive processes, causation) for suicidal behavior; past suicide attempt predicted best by sentiment (positive tone, joy, anticipation,

sadness), and current suicidal ideation by function words (personal pronouns, quantity words, verbs, adverbs, second person pronouns, determiners, adjectives; McBride et al., 2025).

Synthesis of psychiatric disorders research. NLP can be integrated into multimodal AI frameworks for psychiatry (e.g. bipolar disorder) for population-level screening, refining diagnosis for ambiguous symptoms, stratifying patients into subgroups, and assembling homogeneous research cohorts (Smeland et al., 2025). These models synthesize diverse data into actionable precision-medicine insights that can be applied across the healthcare continuum. For neuropsychologists, these applications offer the potential for more objective symptom tracking, risk assessment, and personalized treatment approaches in psychiatric populations. However, this is the area where the gap between research and clinical practice is widest. Key questions remain regarding pathognomonic markers, generalizability, sampling frequency, non-help-seeking populations, differentiating aggression types, suicide risk accuracy, and integrating data into a clinical decision support system (CDSS) using candidate markers validated with clinical rating scales and functional correlates initially with human-in-the-loop iterative technologies (Chandler et al., 2022).

Challenges and recommendations for future directions in NLP in clinical neuropsychology

Limitations of the reviewed studies

Across the diverse clinical domains reviewed above, NLP demonstrates consistent and generally adequate accuracy in classification and detection tasks within specific research contexts. NLP approaches have advanced to instruments capable of augmenting routine neuropsychological practice. When built on transparent pipelines that combine annotated speech, EHR narratives, and psychometric metadata, NLP offers temporal sensitivity, capturing microlinguistic changes before they are observable on standard tests, and can be deployed at scale through telehealth platforms. However, limitations of the reviewed studies were evident, synthesized into key themes: lack of reporting standards leading to translatability and scalability problems; data quality, representativeness and biases; the knowledge gap amongst clinical neuropsychologists and the need for foundational literacy in computational methods; ethical and privacy considerations; and navigating conflicts between clinical intuition and algorithmic output (i.e. clinician-in-the-loop frameworks). While the potential for early detection and prediction is compelling, particularly in psychiatry, these applications carry high ethical stakes and are furthest from clinical implementation, pointing to a translation gap. The field must now pivot from demonstrating proof-of-concept to the more difficult work of clinical validation, establishing generalizability in diverse populations, and ensuring that these tools are transparent, equitable, and safely integrated into clinical workflows.

Current state of adoption and available tools

A variety of tools are available to researchers and healthcare organizations to implement NLP solutions. Open-source statistical software libraries are the most readily

accessible and affordable. For those with technical expertise, open-source Python libraries are popular choices. The Natural Language Toolkit (NLTK) is a foundational library for many basic NLP tasks. SpaCy is known for its speed and efficiency, making it suitable for production environments. For machine learning and deep learning, scikit-learn, TensorFlow, and PyTorch are the standard libraries. Specialized clinical NLP toolkits are available open-source and specifically designed for the clinical domain, containing pre-trained models for medical terminology. These include cTAKES (clinical Text Analysis and Knowledge Extraction System) from the Apache Software Foundation and CLAMP (Clinical Language Annotation, Modeling, and Processing Toolkit). Commercial cloud platforms for major technology companies, which offer powerful, user-friendly NLP services tailored for healthcare. Google Cloud provides Healthcare Natural Language API, which can extract medical information from text. Amazon Web Services offers Amazon Comprehend Medical, which uses NER to identify medical information and is HIPAA-eligible. Microsoft Azure has Text Analytics for health, which performs similar functions. Commercial NLP companies such as John Snow Labs offer a suite of tools, including the popular Spark NLP for Healthcare, which provides state-of-the-art pre-trained models for a wide range of clinical and biomedical NLP tasks. IQVIA also provides an NLP platform to support clinical development and research. While more readily available, affordability then becomes a concern. In research settings, academic institutions have independently developed tools. For example, CognoSpeak is a fully automated tool using automatic speech recognition which classifies patients into MCI, AD, functional cognitive disorder, and healthy controls (O'Malley et al., 2020). An NLP-powered annotation tool (NAT) was developed at Mass General Brigham healthcare to extract and summarize clinical notes including a web tool to present an integrated view of all data (Noori et al., 2022). The Toolkit to Examine Lifelike Language (TELL v2.0) has been deployed across research settings to identify neurodegenerative syndromes through automated speech analysis (García et al., 2025). However, these research tools are not widely disseminated and often not practical for neuropsychologists in private practice. Overall, there is a translation gap in which many tools are not available across neuropsychology practice settings, and knowledge gaps wherein neuropsychologists are unfamiliar with the methods of analysis. The following section outlines recommendations for future research and establishing NLP competencies needed for the evolving role of the neuropsychologist, including current challenges and limitations which should be addressed as NLP adoption increases.

Reporting standards

In global mental health settings, low-cost speech analytics have begun to bridge workforce gaps, facilitating early identification and triage of cognitive impairment in under-resourced regions (García et al., 2023). Yet, methodological heterogeneity necessitates cautious interpretation (Shankar et al., 2025). Future consortia-based initiatives should adopt reporting standards akin to Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis—Artificial Intelligence (TRIPOD-AI; Collins et al., 2024) and include calibration analyses, decision-curve evaluation, and head-to-head comparisons with established neuropsychological measures to solidify the incremental clinical utility of NLP. Reference standards are also

under-reported, and many studies rely on diagnostic codes. To validate NLP markers, comparison to other biomarkers is recommended. Agreement on diagnostic precision cutoffs for clinical trials is necessary, for example an AUC 0.70–0.80 is considered acceptable, 0.80–0.90 excellent, and AUC > 0.90 outstanding (Hosmer et al., 2013). While some studies did report inter-rater agreements, few have systematically evaluated incremental validity of NLP over conventional tests or other objective biomarkers, and this should be included in reporting standards. A further methodological challenge is the high-dimensional nature of linguistic data. Studies often extract hundreds of features, increasing the risk of spurious correlations and false positives if appropriate corrections for multiple comparisons are not applied. Future research should more consistently report effect sizes and utilize validation strategies like nested cross-validation to ensure that findings are robust and not simply the result of chance. Finally, low model accuracy can stem from numerous factors, including small or homogenous training datasets that fail to capture the diversity of the clinical population, poor quality of input data (e.g. noisy audio recordings), or a fundamental mismatch between the linguistic features being measured and the clinical outcome being predicted. Therefore, when appraising a study, neuropsychologists should pay close attention to the characteristics of the training and validation samples, the definition and reliability of the "ground truth" used to train the model, and the transparency of the performance reporting.

Data representativeness, and bias

Many training data using speech corpora remain skewed toward highly educated, English-speaking, and predominantly White cohorts, raising concerns regarding algorithmic bias and diagnostic inequity. Future studies should prospectively stratify enrolment across age, education, language background, and cultural context, and report error rates within each subgroup. Practitioners must evaluate whether performance metrics are stratified and document any disparities in a report limitations section. Supervisors should model the ethical imperative to obtain informed consent for secondary language-data use, particularly in vulnerable populations. Cultural and linguistic sensitivity is a related area requiring further development. Effective use of NLP in diverse populations requires awareness of dialectal variation, codeswitching, and culturally bound discourse styles (García et al., 2023). Practitioners should consult with bilingual colleagues and consider local corpora or domain-adapted models to avoid misclassification, thereby upholding linguistic justice principles. Any predictive model must be trained on data that are broad, representative, and collected in a standardized way, to avoid amplifying existing healthcare inequities (Romano et al., 2023). Sampling bias is a key concern because the clinical expression of many neurological conditions differs by age, sex, ethnicity, and a range of social determinants of health. For example, for Alzheimer's disease, an LLM built from records of adults over 65 may perform poorly when asked to flag early-onset cases. Approaches such as oversampling underrepresented populations and applying debiasing algorithms during training can lessen these distortions. Similar principles apply to sentiment analysis in clinical notes: misreading emotional tone can lead to misguided interventions. Robust performance therefore demands exposure to a wide range of linguistic

patterns and culturally specific expressions of emotion. Incorporating clinician feedback loops to review and adjust the model's sentiment judgments may improve accuracy and safety. Conversely, NLP itself could be used to identify language barriers and cultural context from unstructured EHR data (Martin et al., 2024). Addressing these issues is important in ensuring that NLP tools enhance, rather than detract from, equitable access to neuropsychological services. Further, the uptake of NLP may increase access to early identification and reduce possible bias from subjective assessment methods.

Interpretability and explainable AI

Clinician adoption hinges on models that not only perform well but also provide interpretable outputs. Interpretability tools should be developed that allow clinicians to see how the model arrived at a given recommendation, improving confidence and uptake. Posthoc explanation frameworks, such as SHapley Additive exPlanations (SHAP) and integrated-gradient visualizations, can be applied to highlight salient lexical features driving diagnostic decisions, fostering trust and facilitating clinical auditing (Crema et al., 2022). Embedding these explainers within graphical dashboards allows neuropsychologists to cross-reference linguistic cues with behavioral observations, aligning machine inferences with domain intuition. A competency for neuropsychologists may include the ability to translate model explanations (e.g. feature importance plots, saliency maps) into clinically meaningful insights. Trainees should practice reconciling machine-identified linguistic markers with behavioral observations during feedback sessions, thereby fostering a transparent clinician–patient alliance. Another primary gap in the current understanding is the precise mapping of specific NLP-derived linguistic features onto discrete neuropsychological constructs. For instance, while a model may detect changes in verbal fluency, further work is needed to determine if these changes reflect a decline in executive functions, lexical access, or processing speed. Similarly, longitudinal studies are needed to understand how these digital biomarkers evolve with disease progression or respond to treatment.

Foundational literacy in NLP concepts

Clinicians should demonstrate fluency with key terminology (e.g. tokenization, embeddings, transfer learning) and appreciate how different model architectures (e.g. BERT versus GPT) influence interpretability and performance (Shakeri & Farmanbar, 2025). There is a knowledge gap amongst neuropsychologists which can best be addressed through modification of training programs and offering ongoing continued professional development courses. Training programs could integrate structured teaching on these topics into didactics, ensuring that trainees can critically appraise emerging literature and communicate algorithmic outputs in plain language to patients and colleagues. Clinicians should be able to match a clinical question to an appropriate NLP task (e.g. named entity recognition for automated chart review, transformer-based text classification for early dementia screening). Skills also include examining internal and external validation procedures, ensuring that calibration statistics and decision-curve analyses meet psychometric standards before adopting a model in practice. Embedding this competency framework within training curricula and credentialing standards will

ensure that the next generation of neuropsychologists can harness NLP ethically, effectively, and equitably, thereby fulfilling the transformative promise outlined in this review and solidifying their role as experts in traditional and technologically-augmented cognitive assessment. Supervisors should encourage participation in interdisciplinary journal clubs, hackathons, and open-source benchmarking initiatives, embedding a culture of perpetual skill renewal. Collaborations and cross-disciplinary workshops should be encouraged in training programs, and membership within organizations such as the Association for Computational Linguistics. Neuropsychology-specific conferences could encourage presentations and interactive workshops on computational and statistical methods.

Ethical and privacy considerations

Language data routinely contain sensitive personal and health information. Compliant deidentification pipelines that blend rule-based redaction with contextual named entity recognition are essential to meet Health Insurance Portability and Accountability Act (HIPAA) and General Data Protection Regulation (GDPR) mandates (Clay et al., 2025). LLM deployments must additionally guard against inadvertent memorization of protected health information and implement consent processes that are accessible to cognitively-impaired individuals (Romano et al., 2023). Practitioners must implement HIPAA-compliant deidentification pipelines and understand the specific obligations imposed by GDPR when serving international patients. Competence extends to recognizing the risks of data leakage and model memorization, and to employing privacy-preserving techniques such as differential privacy or federated learning when indicated (Romano et al., 2023). There is also suggestion that OpenAI poses a threat to performance validity test security, although ethical opposition (i.e. LLMs refusing to answer questions which may violate ethical principles) has increased over time (Lavigne et al., 2024). Navigating these complex ethical landscapes is a critical responsibility for neuropsychologists utilizing NLP. Concrete suggestions for neuropsychologists to take today include: (1) Verifying that any third-party NLP tool has a signed Business Associate Agreement (BAA) to ensure HIPAA compliance; (2) Using only de-identified data for any local research or model training projects, employing robust tools for scrubbing PHI; (3) Ensuring patient consent forms for clinical services and research are updated to explicitly include the potential use of their de-identified language data for quality improvement or research purposes; and (4) Advocating within institutions for clear policies on the governance and storage of sensitive language data. Finally, to facilitate the integration of biomarkers into primary care, policy reform must establish national quality measures for the identification and management of cognitive impairment, aligning clinical and financial incentives to encourage health systems to prioritize early diagnosis, including engaging key organizations (e.g. United States Preventive Services Taskforce, Centers for Medicare and Medicaid Services, the Agency for Healthcare Research and Quality) to develop and implement these standards (Borson et al., 2025).

Integration into clinical neuropsychology workflows

Effective implementation demands alignment with existing documentation practices and reimbursement structures. Embedding NLP modules directly into electronic note

templates can automate differential diagnosis checklists and populate semi-structured impression fields, potentially reducing clinician workload. Seamless integration with voice-recognition platforms further enables real-time feedback during evaluations, guiding examiners toward probing questions when narrative complexity dips below normative thresholds (Clay et al., 2025). Ambient listening technologies (ALTs) offer real-time transcription and EHR integration, reducing documentation burden and allowing clinicians to focus on patient interaction (Stults et al., 2025). NLP-augmented ALTs could enhance efficiency, data accuracy, add unique data to current assessment practices, and improve care quality. Some EHRs (Epic) use ALT, but uptake in neuropsychology is limited. Recent work has demonstrated the utility of ambient intelligence by embedded DAX Copilot in Epic for transcription, seizure forecasting and prevention, and clinical decision support, although it also noted difficulties with fabrications and omissions (Kakwan et al., 2025). Integrating NLP tools effectively will require collaboration and adaptation of existing clinical practices, ultimately freeing up clinician time for higher-level cognitive tasks and patient-centered care. Further, reports should explicitly state the NLP tool, version, validation provenance, and interpretive boundaries, mirroring best practice for psychometric test description. Given the pace of NLP innovation, clinicians must engage in ongoing dialogue with data scientists, speech-language pathologists (SLPs), and ethicists.

The clinician-in-the-loop: navigating algorithmic and human expertise

A key challenge arises when a model's output conflicts with a clinician's judgment. In such cases, the tool should not be seen as an arbiter of truth, but as a source of additional data. For instance, the most rapid adoption of NLP methods in neuropsychology has been use of LLMs as these tools have become more readily accessible. Studies comparing AI diagnostic accuracy with human clinicians across medical fields show variable performance. A review of 83 studies found overall AI accuracy was 52.1%, not significantly different from non-expert physicians but worse than expert physicians (Takita et al., 2025). However, performance of later models (GPT-4, GPT-4o, Llama3 70B, Gemini 1.0 Pro, Gemini 1.5 Pro, Claude 3 Sonnet, Claude 3 Opus, Perplexity) improved compared to earlier versions (GPT-4, GPT-3; Takita et al., 2025). Further, a two-phase study assessed ChatGPT-4 Omni's utility for neuropsychological differential diagnosis, in Phase 1 finding only modest concordance between ChatGPT's few-shot outputs and clinician diagnoses (Cohen's $\kappa = 0.33$), despite excellent internal consistency ($\alpha = 0.96$) and heavy reliance on Montreal Cognitive Assessment (MoCA) and episodic-memory scores, and Phase 2 found no statistically significant alignment between zero-shot ChatGPT variants and expert judgment (Dimmick et al., 2025). These findings highlight both the promise and limitations of LLMs in neuropsychological triage; while approximating clinician decisions under highly structured prompting, accuracy is below the threshold required for independent diagnostic use, emphasizing the necessity of human oversight in cognitive-disorder assessment. The clinician's role is to critically evaluate the discrepancy, whether the model is flagging a subtle linguistic feature missed during interview and assessment, or if the model's recommendation is based on a spurious correlation from the training data that does not apply to the specific, complex patient. This requires clinicians to have the

competency to "audit" the AI's suggestion, using explainability tools (e.g. SHAP) to understand the "why" behind the prediction. The ultimate decision must rest on an integrated formulation that synthesizes the clinician's expertise, the patient's context, traditional test data, and the NLP-derived insights. In the current state of the field, NLP models can be conceptualized as consultants which require a "clinician-in-the-loop," rather than the final authority.

Implementation in patient feedback

While independent diagnosis is far from ready for clinical primetime in neuropsychology, an application ready for adoption involves neuropsychologists co-creating feedback with patients, using visual dashboards to explain linguistic metrics and linking recommendations to personalized rehabilitation goals. This human-centered approach to interpreting and communicating AI findings is a unique strength of the neuropsychology profession. In a proof-of-concept study by the author (BW), the application of NLP and LLMs was explored to generate personalized neuropsychological feedback and recommendations, offering a scalable approach to clinical practice. It used 114 de-identified neuropsychological reports with clinician-written recommendation sections removed. An NLP pipeline was created using Python and the spaCy library for data extraction, followed by feeding de-identified report segments into OpenAI's GPT-4 to generate recommendations. Subsequent qualitative thematic analysis, facilitated by NVivo, and cross-validation in R (using tidytext and topicmodels) categorized the feedback. The LLM generated 877 unique neuropsychological recommendations. Comparative analysis revealed distinct patterns; for instance, reports for patients with depressive disorders predominantly included behavioral interventions and cognitive restructuring, while those for neurodegenerative conditions focused on cognitive rehabilitation and caregiver support. All feedback consistently included "actions," "resources," and a "timeline." However, 412 (49.2%) of the 877 recommendations may have been overly prescriptive or outside of scope for neuropsychologists. Thematic analysis identified key recommendation categories such as cognitive rehabilitation strategies, behavioral interventions, lifestyle modifications, home and environment modifications, planning for future care, and pharmacological suggestions. Given the simplicity of this approach and ease of replicability, this points to the potential for AI to augment clinical neuropsychological recommendation processes during feedback sessions, which could be incorporated into standard training.

Longitudinal and multimodal monitoring

For neuropsychologists employing EHR to track patient progress, training may include establishing longitudinal dashboards that flag deviation from baseline narrative complexity, allowing proactive care adjustments. Longitudinal modeling of cognitive sequelae of ischemic stroke, brain tumors, cancer-related cognitive impairment, global atrophy, aphasia, and remission monitoring is possible. For instance, the Cookie Theft picture description could be employed at routine evaluations where a clinician scores for content units, while an NLP tool simultaneously analyzes the patient's output for acoustic features (e.g. pause duration, pitch variability), syntactic complexity (e.g. clause density), and semantic coherence (e.g. topic drift). These metrics, captured and

tracked longitudinally in a dashboard, could offer greater sensitivity to subtle cognitive decline than a simple score, flagging changes in verbal fluency or organization months before they are apparent on standard testing. Recurrent neural networks (RNNs) and other temporal deep learning models can be used for analyzing serial clinical and imaging data to chart disease trajectories, estimating prognosis, and uncovering early biomarkers across diverse neurological populations (Ying et al., 2024). Researchers could consider combining traditional tests with wearable-sensor data to enhance ecological validity and obtain multiple sources of linguistic data for NLP analysis, with the goal of including multimodal analyses as part of clinical workflows. There is already uptake in computational neuroscience, neuroimaging, and cognitive data (Ying et al., 2024). Examples include electroencephalography (EEG) pattern analysis for seizure detection, dynamical-systems simulations and Bayesian decision models, brain morphology, and supervised algorithms translating diffusion tensor imaging (DTI) findings of white matter integrity and voxel-based morphometry (VBM) for grey matter density into predictions of language or memory outcome (Ying et al., 2024). These integrated approaches represent a step forward in understanding brain-behavior relationships and predicting clinical outcomes.

Conclusion

Natural language processing has crossed the threshold from experimental novelty to clinically actionable technology with implications for the field of clinical neuropsychology. By providing objective, quantifiable data on spoken and written language and communication, NLP can augment traditional neuropsychological assessment methods, enhancing diagnostic precision, improving efficiency, and supporting personalized interventions across a range of neurocognitive and psychiatric conditions. However, to realize this potential, the field must move forward with a clear understanding of its current limitations and the role neuropsychologists must play in shaping its future. As domain experts, neuropsychologists are uniquely positioned to ensure that NLP research asks clinically relevant questions, that assessment tasks possess ecological validity, and that model outputs are interpreted through the lens of established cognitive theory. They are the bridge between computational methods and patient-centered application. To harness its full potential and ensure responsible integration into practice, future endeavors must prioritize addressing current limitations: (1) recruiting diverse cohorts and publishing debiased error analyses to address algorithmic bias; (2) adopting explainable AI frameworks for every production model to foster trust and interpretability; (3) implementing privacy-preserving pipelines to protect sensitive patient data; (4) embedding NLP within electronic health record ecosystems to facilitate widespread adoption; and (5) committing to open benchmarks and transparent reporting to promote reproducibility and collaboration, including formally assessing the incremental value of NLP in clinical neuropsychology. Concerted progress along these axes promises to elevate neuropsychological assessment, making it more objective, sensitive, and equitable for the populations it serves. Embracing NLP and developing necessary competencies for use will be essential for clinical neuropsychologists to remain at the forefront of cognitive assessment in the digital age.

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Appendix A: Search strategy and results

Eligibility criteria

Participants/population

Studies involving patients who underwent clinical neuropsychological assessment, broadly defined as the evaluation of cognitive function (e.g. memory, attention, language, executive function) in a clinical or research setting, or observational studies on individuals with neuropsychological conditions, or syndromes/conditions involving neurocognitive dysfunction, including psychiatric disorders with a cognitive component (e.g. schizophrenia).

Intervention/exposure

Studies were required to use a natural language processing (NLP) method (e.g. text mining, computational linguistics) to analyze clinical data, typically patient responses, medical notes, or transcripts of neuropsychological interviews. As "NLP" is not a universally applied term in relevant studies, other terms included: computational linguistics, text mining, text analysis, speech analysis, language model, transformer, machine learning or deep learning (as applied to clinical text or speech), and artificial intelligence (as applied to clinical text or speech).

Comparators

Any standard neuropsychological assessment, diagnostic procedure, or classification tool. However, due to the limited number of studies, comparison studies were optionally included in the search criteria. Where applied, NLP-based methods were compared to conventional approaches (e.g. standard clinical or neuropsychological assessments).

Outcomes

Studies evaluating the diagnostic accuracy, reliability, validity, feasibility, or potential clinical utility (e.g. diagnostic accuracy, predictive value) of NLP for neuropsychological assessment or intervention.

Study designs

Study designs included randomized controlled trials (RCTs), quasi-experimental studies, observational studies (cohort, cross-sectional, case-control), and relevant qualitative or mixed-methods studies. Conference abstracts, editorials, or narrative reviews were excluded unless they presented original data.

Language/date restrictions

Only articles in English were included. No publication date restriction was applied.

Information sources

Scoping search of the following electronic databases: PubMed (MEDLINE); Embase; PsycINFO; Scopus; Web of Science (or Elsevier Scopus); Google Scholar; CINAHL; Cochrane Register of Controlled Trials (CENTRAL). In addition, the reference lists and new citing articles of selected eligible full-text articles were scanned to identify any additional studies. The main database search was conducted on 15 May 2025, with a broad review for interval publications conducted on 1 July 2025.

Search strategy

A search strategy was developed to include both free-text keywords and database-specific controlled vocabulary (e.g. MeSH terms in PubMed). Below is an example of a combined (Boolean) search string adapted for PubMed; similar concepts and keywords were used for other databases, with adjustments to syntax as per search requirements.

("natural language process*" OR NLP OR "computational linguistics" OR "text mining" OR "text analysis" OR "machine learning" OR "language model" OR "transformer")AND ("clinical neuropsycholog*" OR neuropsycholog* OR "cognitive function" OR "cognitive assessment" OR "cognitive disorder" OR "neurocognitive disorder" OR "dementia" OR "Alzheimer*" OR "Parkinson*")AND (assess* OR diagnos* OR evaluation OR screening OR intervention OR treatment)\text{("natural language process*" OR NLP OR "computational linguistics" OR "text mining" OR "text analysis" OR "machine learning" OR "language model" OR "transformer")}\\\text{AND ("clinical neuropsycholog*" OR neuropsycholog* OR "cognitive function" OR "cognitive assessment" OR "cognitive disorder" OR "neurocognitive disorder" OR "dementia" OR "Alzheimer*" OR "Parkinson*")}\\\text{AND (assess* OR diagnos* OR evaluation OR screening OR intervention OR treatment)}("natural language process*" OR NLP OR "computational linguistics" OR "text mining" OR "text analysis" OR "machine learning" OR "language model" OR "transformer")AND ("clinical neuropsycholog*" OR neuropsycholog* OR "cognitive function" OR "cognitive assessment" OR "cognitive disorder" OR "neurocognitive disorder" OR "dementia" OR "Alzheimer*" OR "Parkinson*")AND (assess* OR diagnos* OR evaluation OR screening OR intervention OR treatment)